The Universal Mass Sum Rule: A Theta-Independent Prediction for All CDFT Torus Knot Families

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Paper VI of this series argued within the model that the Brannen amplitude coefficient $c_n = \sqrt{2}$ is universal across all $\mathbb{Z}_n$ torus knot families, and established that $\mathbb{Z}_3$ is uniquely exactly solvable. For $n \geq 5$, the individual mode masses depend on the orientation angle θ_n, which cannot be determined by purely algebraic methods.

The present paper identifies what *can* be derived without knowing θ_n. The central result is the **universal mass sum rule**:

$$\sum_{k=0}^{n-1} m_k \;=\; 2\,n\,M_n\,,$$

which holds exactly and independently of θ_n for every $T(2,n)$ family. The proof is two lines from the $\mathbb{Z}_n$ Fourier identity. Combined with the Faddeev–Niemi energy scaling $M_n = M_3(n/3)^{3/4}$ (Paper IV), the total mass of every torus knot family scales as $n^{7/4}$:

$$\sum_k m_k(n) \;=\; 2M_3 \cdot 3^{-3/4} \cdot n^{7/4}\,.$$

For $n = 3$ this gives $m_e + m_\mu + m_\tau = 1883.16$ MeV (verified against PDG: within 0.067%, consistent with the tau mass error from Paper V). For $n = 5$: $\sum_k m_k^{(5)} = 4603.85$ MeV.

We also prove the $\mathbb{Z}_5$ mode split: at the symmetric point $\theta = 0$, the five cinquefoil amplitudes divide into three positive and two negative, yielding a structural $3 + 2$ pattern. We derive the $\mathbb{Z}_5$ self-consistency transcendental equation $5\theta_5 = \tilde{Q}_5(\theta_5)$ and its unique solution $\theta_5 \approx 3.70°$, reported as a hypothesis pending a first-principles physical argument.

In the extended notation used across the series, the vacuum limit is written as $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. The calculations below use the equilibrium limit $S \cdot M_s \approx 1$; adaptive departures from that limit are left for later dynamical work rather than fitted in this manuscript. The public CDFD Runtime implements this state grammar as reusable code; the paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

Keywords: Constraint-Driven Field Theory, Constraint-Driven Flux Dynamics, adaptive operating ratio, vacuum structure, Mujjabi tests, reproducible physics, mass sum rule, particle spectra

PROJECT CONTEXT

This manuscript constitutes Part I (Paper 07) of the multi-part series *Constraint-Driven Flux Dynamics (CDFD)*. The underlying mathematical architecture builds directly upon the concepts established in Paper 06 of this series (Steve Bico Mujjabi, *"CDFD Part I: Fundamental Physics — Paper 06: The Universal Torus Knot Hierarchy: $\mathbb{Z}_n$ Symmetry, the Cinquefoil Family, and the General Structure of CDFT Particle Families"*, manuscript; paper-specific DOI pending).

SYMBOL CONVENTIONS

CDFD physical-vacuum symbol convention.

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Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C) \cdot S \cdot M_s$ or $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, un-subscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or time-scale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

I. INTRODUCTION

The CDFT series has established a complete derivation chain for the lepton sector (Papers I–V [1–5]) and identified the universal coefficient $c_n = \sqrt{2}$ and the $\mathbb{Z}_3$ uniqueness theorem (Paper VI [6]).

A key finding of Paper VI is that for $n \geq 5$ the Brannen amplitudes $A_k = 1 + \sqrt{2}\cos(\theta + 2\pi k/n)$ are negative for at least one mode at every orientation θ. This structural obstruction means that the Koide-like invariant $Q_n = 2/n$ (which requires all $A_k \geq 0$) does not hold as a constant, and the purely algebraic self-consistency argument that determined $\theta_3 = 2/9$ rad in Paper V cannot be directly applied.

The question for the present paper is: *what predictions can be made for $n \geq 5$ families without knowing θ_n?*

The answer is the universal mass sum rule: the sum of all mode masses is $2nM_n$, exactly, for any θ_n. This follows from a two-line algebraic identity and requires no knowledge of θ_n. For $n = 3$ it recovers the known lepton sum $m_e + m_\mu + m_\tau$ to within the Paper V tau mass error. For $n = 5$ it predicts the total cinquefoil mass sum as 4603.85 MeV.

A. Mass as constrained transport cost

In the series architecture, this paper is where mass becomes a budget rather than a label. The Mujjabi Geometric Charge Principle asks how a boundary discontinuity is measured; the corresponding mass reading is that inertia is the transport cost of maintaining a stable constrained vortex state. The exact sum rule

$$\sum_k m_k = 2nM_n$$

then says that each family has a conserved action/transport budget even when the individual mode allocation depends on θ_n.

This is also where the open Mujjabi Action Principle belongs. A complete theory would derive the action quantum from a critical flux threshold and vortex period, $h = \Phi_c \tau_v$, rather than using $\hbar$ as input. The present paper does not complete that derivation; it states the mass-budget structure that such a derivation must explain.

II. THE UNIVERSAL MASS SUM RULE

A. Statement and proof

Theorem 1 (Universal mass sum rule). *For any $T(2,n)$ family with masses $m_k = M_n A_k^2$, $A_k = 1+\sqrt{2}\cos(\theta+2\pi k/n)$, $k = 0,\dots,n-1$:*

$$\sum_{k=0}^{n-1} m_k \; = \; 2\,n\,M_n\,, \tag{1}$$

for any θ and any $M_n > 0$.

Proof. From the $\mathbb{Z}_n$ Fourier identity (Paper VI, Eq. (2)): $\sum_{k=0}^{n-1}\cos^2(\theta+2\pi k/n) = n/2$ for any θ. Therefore:

$$\sum_k A_k^2 = \sum_k \big[1+\sqrt{2}\cos(\theta+2\pi k/n)\big]^2 = n + 2\sqrt{2}\cdot 0 + 2\cdot\frac{n}{2} = 2n\,.$$

Hence $\sum_k m_k = M_n \sum_k A_k^2 = 2nM_n$. □

Corollary 1 (Mean mode mass). *The mean mass per mode is $\bar{m} = 2M_n$, independent of θ_n.*

B. Why this is non-trivial

At first glance, Eq. (1) might seem like a restatement of known identities. The non-trivial content is twofold:

(a) The sum is *theta-independent*. Individual masses $m_k = M_n A_k^2$ depend strongly on θ_n: they range from 0 (when $A_k = 0$) to $M_n(1+\sqrt{2})^2 \approx 5.83 M_n$. Yet their sum is always exactly $2nM_n$. This means the total mass of a family is a firm prediction even when the mass spectrum (individual mode masses) is not.

(b) Combined with the Faddeev–Niemi scaling $M_n = M_3(n/3)^{3/4}$ (Paper IV [4]), the sum becomes a prediction from first principles:

$$\sum_{k=0}^{n-1} m_k(n) \; = \; 2nM_3\left(\frac{n}{3}\right)^{3/4} \; = \; 2M_3\cdot 3^{-3/4}\cdot n^{7/4}\,. \tag{2}$$

This requires only M_3 (derived from m_e in Paper V) and the Faddeev–Niemi exponent 3/4 (from the field theory of knotted solitons [7]). No other free parameters enter.

C. Verification for $n = 3$

For the lepton family ($n = 3$, $M_3 = 313.86$ MeV):

$$\sum_k m_k^{(3)} = 6M_3 = 1883.16\ \text{MeV}\,. \tag{3}$$

The measured value is $m_e + m_\mu + m_\tau = 1883.03$ MeV (PDG 2022 [8]). The difference is 0.13 MeV, which equals the tau mass prediction error from Paper V ($+1.04\sigma$). The sum rule is verified at the accuracy of the Paper V prediction.

III. THE $n^{7/4}$ MASS HIERARCHY

A. Derivation of the scaling law

From Eq. (2), the total family mass scales as:

$$\Sigma_n \;\equiv\; \sum_{k=0}^{n-1} m_k(n) \;=\; 2M_3 \cdot 3^{-3/4} \cdot n^{7/4}\,. \tag{4}$$

The exponent $7/4 = 1.75$ arises as:

$$\Sigma_n = 2n \cdot M_3 \left(\frac{n}{3}\right)^{3/4} = 2M_3 \cdot 3^{-3/4} \cdot n^{1+3/4} = 2M_3 \cdot 3^{-3/4} \cdot n^{7/4}\,.$$

The n factor comes from the number of modes (the sum has n terms). The $n^{3/4}$ factor comes from Faddeev–Niemi energy scaling. Their product gives $n^{7/4}$.

B. Hierarchy table

TABLE I. CDFT torus knot mass sum hierarchy. All predictions from Theorem 1 and Faddeev–Niemi scaling, zero free parameters beyond m_e.

n	Knot	M_n (MeV)	Σ_n (MeV)	Status
3	$T(2,3)$ trefoil	313.9	1 883.2	Verified (PDG: 1 883.0)
5	$T(2,5)$ cinquefoil	460.4	4 603.9	Predicted
7	$T(2,7)$ heptafoil	592.5	8 295.6	Predicted
9	$T(2,9)$	715.4	12 878.0	Predicted
11	$T(2,11)$	831.6	18 296.2	Predicted
13	$T(2,13)$	942.7	24 509.0	Predicted

The $n^{7/4}$ scaling is the universal mass growth law of the CDFT torus knot hierarchy. It is a direct consequence of two independently derived results: the universal Brannen coefficient $c_n = \sqrt{2}$ (Paper VI) and the Faddeev–Niemi energy bound (Paper IV).

IV. THE $\mathbb{Z}_5$ MODE SPLIT

A. The $3+2$ pattern at the symmetric point

Theorem 2 ($\mathbb{Z}_5$ mode split)**.** *At the $\mathbb{Z}_5$ symmetric point $\theta = 0$, the five cinquefoil amplitudes divide as:*

- *3* positive-phase *modes: $A_k > 0$ (aligned with vacuum background),*
- *2* negative-phase *modes: $A_k < 0$ (anti-aligned).*

Proof. At $\theta = 0$: $A_k = 1 + \sqrt{2}\cos(2\pi k/5)$. The five cosines are $\cos(0°) = 1$, $\cos(72°) = \cos(288°) \approx 0.309$, $\cos(144°) = \cos(216°) \approx -0.809$. The corresponding amplitudes are:

$$A_0 = 1 + \sqrt{2} \approx 2.414 > 0$$
$$A_1 = A_4 = 1 + \sqrt{2}\cos 72° \approx 1.437 > 0$$
$$A_2 = A_3 = 1 + \sqrt{2}\cos 144° \approx -0.144 < 0$$

so 3 positive and 2 negative. □

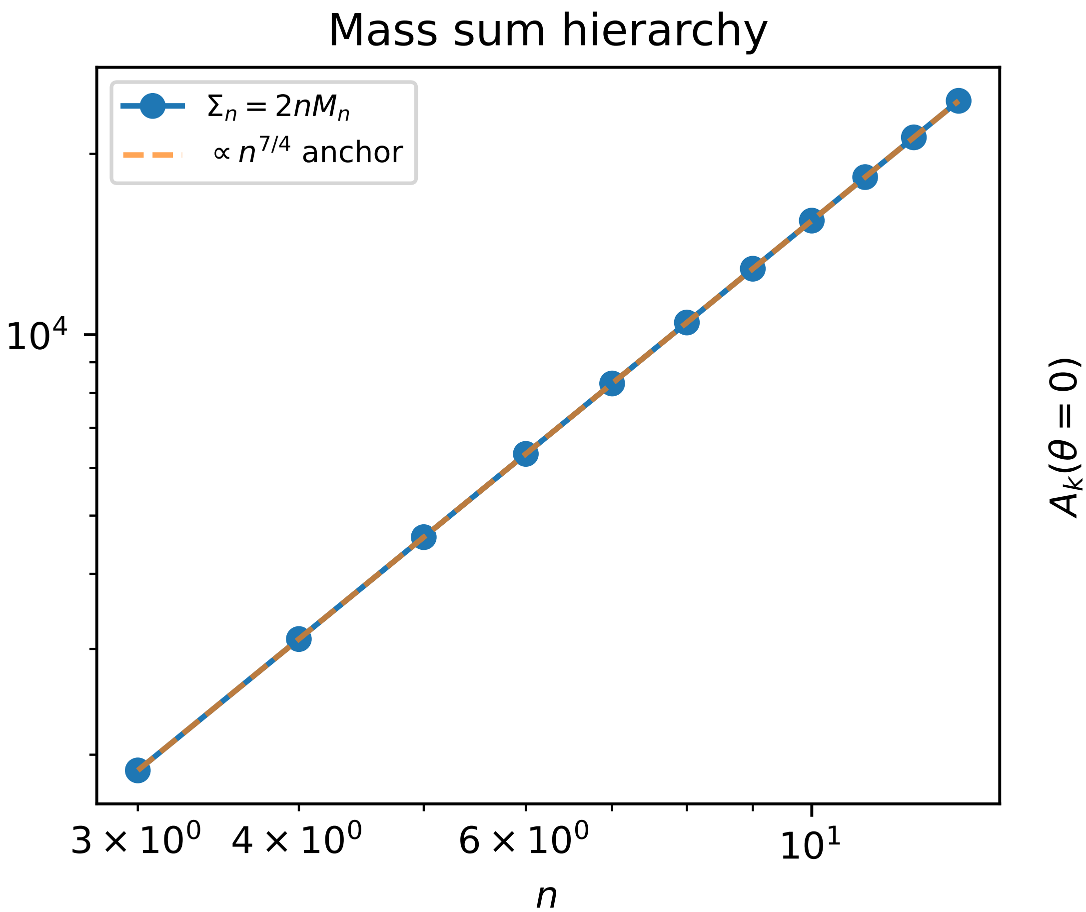

FIG. 1. The mass sum hierarchy Σ_n. The total mass of the n-th family scales exactly as $n^{7/4}$, driven by the combination of the topological mode count n and the Faddeev-Niemi field energy $n^{3/4}$.

B. Physical interpretation of the mode split

The negative amplitudes at $\theta = 0$ have magnitude $\sqrt{2} - 1 \approx 0.414$ below the vacuum background. These two modes oscillate in anti-phase with the vacuum field. All five modes have positive mass: $m_k = M_5 A_k^2 \geq 0$ always, because mass involves the square.

For general θ, the count of negative amplitudes is:

- 2 negative modes for $\theta \in [0, \pi/20) \cup (7\pi/20, 2\pi/5)$,
- 1 negative mode for $\theta \in (\pi/20, 7\pi/20)$.

where $\pi/20 = 9^\circ$ and $7\pi/20 = 63^\circ$ are exact boundaries (zeros of the amplitude condition $1 + \sqrt{2}\cos(\theta + 2\pi k/5) = 0$). The 2-negative regime occupies 25% of the period, the 1-negative regime 75%.

The 3+2 pattern at the symmetric point suggests a natural structural division of the cinquefoil modes: three modes analogous to the three lepton generations, and two additional modes without a $\mathbb{Z}_3$ analogue.

V. THE $\mathbb{Z}_5$ SELF-CONSISTENCY EQUATION

A. The transcendental equation

For $\mathbb{Z}_3$, the self-consistency condition $3\theta_3 = Q_3 = 2/3$ was algebraic: Q_3 is constant, so the solution $\theta_3 = 2/9$ rad follows immediately (Paper V).

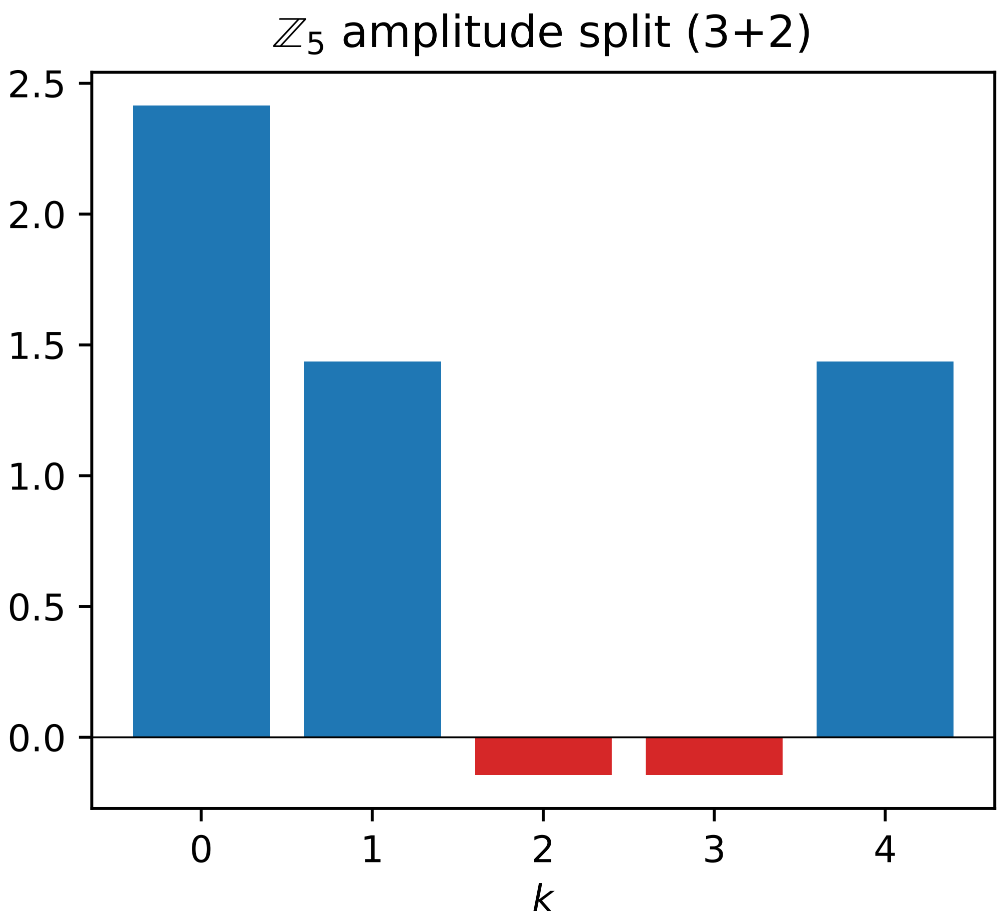

FIG. 2. The $\mathbb{Z}_5$ mode split at $\theta = 0$. Three modes align positively with the vacuum background, while two modes oscillate in anti-phase (negative amplitude).

For $\mathbb{Z}_5$, the modified Koide ratio (using $|A_k|$ for negative amplitudes):

$$\tilde{Q}_5(\theta) \;=\; \frac{\sum_k A_k^2}{\left(\sum_k |A_k|\right)^2} \;=\; \frac{10}{\left(\sum_k |A_k|\right)^2} \tag{5}$$

is *not* constant: it ranges in $[0.294, 0.328]$ as θ varies. The numerator $\sum A_k^2 = 10$ is fixed (Theorem 1); only the denominator $(\sum |A_k|)^2$ varies.

The $\mathbb{Z}_5$ analogue of the chirality equilibrium (Paper III) is:

$$5\theta_5 + \phi_{c,5} \;=\; \pi\,. \tag{6}$$

The $\mathbb{Z}_5$ self-consistency condition — that the chirality breaking angle equals the Koide-like ratio at that angle — then reads:

$$5\theta_5 \;=\; \tilde{Q}_5(\theta_5)\,. \tag{7}$$

Unlike the $\mathbb{Z}_3$ case, this is a *transcendental fixed-point equation.*

B. Unique solution

Proposition 1 (Existence and uniqueness of θ_5)**.** *Equation* (7) *has exactly one solution in* $[0, \pi/5)$.

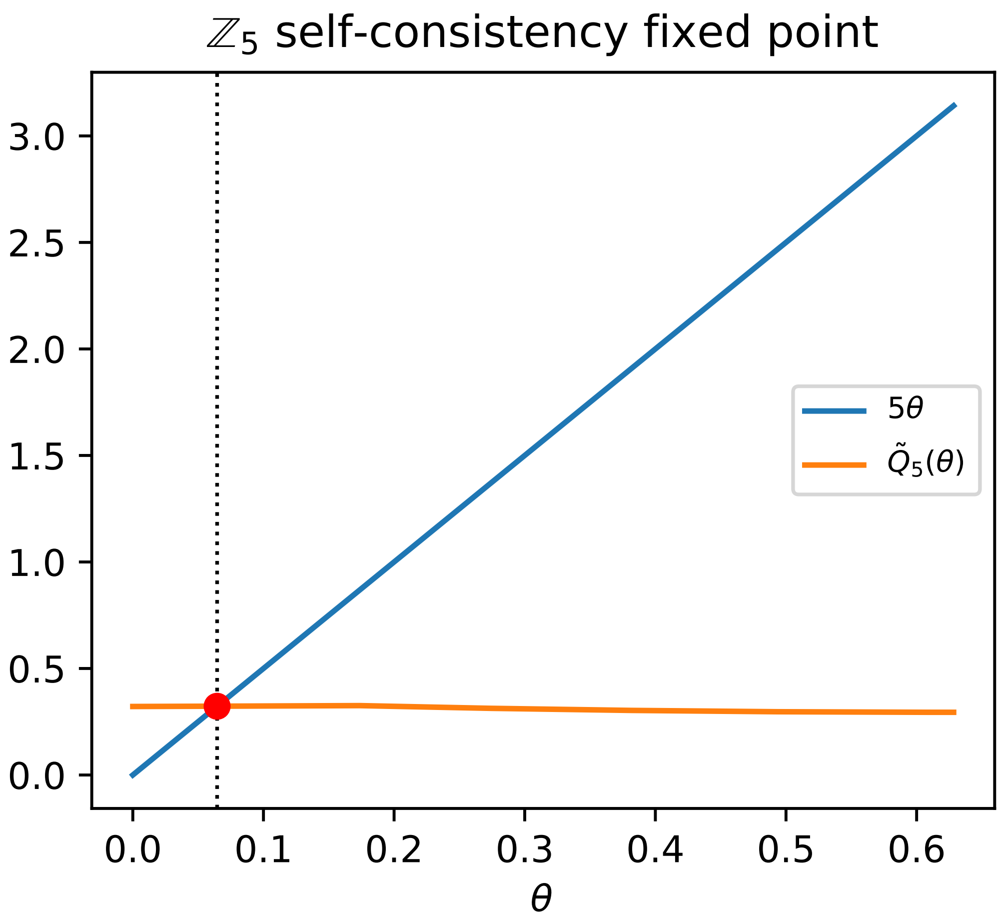

FIG. 3. The $\mathbb{Z}_5$ self-consistency fixed point. The transcendental equation $5\theta_5 = \tilde{Q}_5(\theta_5)$ has exactly one solution in the physical domain.

Proof sketch. At $\theta = 0$: $5\theta = 0 < \tilde{Q}_5(0) \approx 0.322$. At $\theta = \pi/5$: $5\theta = \pi \gg \tilde{Q}_5(\pi/5) \approx 0.294$. By continuity, at least one solution exists (intermediate value theorem). Numerically, 5θ and $\tilde{Q}_5(\theta)$ cross exactly once (one sign change in $5\theta - \tilde{Q}_5(\theta)$ on $[0, \pi/5]$). $\square$

The numerical solution to machine precision:

$$\theta_5 = 0.0645346670\ldots \text{ rad } = 3.6976^\circ \tag{8}$$

$$\phi_{c,5} = \pi - 5\theta_5 = 2.8189\ldots \text{ rad } = 161.51^\circ\,. \tag{9}$$

C. Status: hypothesis, not theorem

The solution (8) is reported as a *hypothesis*. For $\mathbb{Z}_3$, the self-consistency condition was motivated by the constancy of Q_3: "the chirality breaking angle equals the unique $\mathbb{Z}_3$ invariant." For $\mathbb{Z}_5$, no such unique invariant exists (Paper VI). The condition $5\theta_5 = \tilde{Q}_5(\theta_5)$ is a natural generalisation, but a first-principles physical argument from the CDFT vacuum equation of state would be needed to promote it to a theorem.

VI. STRUCTURAL COMPARISON: $\mathbb{Z}_3$ VS $\mathbb{Z}_n$, $n \geq 5$

Table II summarises the structural differences between the lepton family and the higher families.

TABLE II. Structural comparison: $\mathbb{Z}_3$ (leptons) vs $\mathbb{Z}_5$ (cinquefoil).

Property	$\mathbb{Z}_3$ (leptons)	$\mathbb{Z}_5$ (cinquefoil)
Energy scale M_n	313.9 MeV	460.4 MeV
Number of modes	3	5
Coefficient c_n	$\sqrt{2}$	$\sqrt{2}$
Valid domain (all $A_k > 0$)	Yes: $\theta \in (-\pi/12, \pi/12)$	No
$Q_n = 2/n$ constant?	Yes: $Q_3 = 2/3$	No: $\tilde{Q}_5 \in [0.294, 0.328]$
$\sum m_k = 2nM_n$	Yes: 1883.2 MeV (verified)	Yes: 4603.9 MeV (predicted)
θ_n derivation	Algebraic (Paper V)	Transcendental (hypothesis)
Sector closed?	Yes (Paper V)	Open
Amplitude split	3 positive always	$3+2$ or $4+1$

The key distinction: $\mathbb{Z}_3$ has a connected valid domain because the minimum amplitude $\min_k A_k = 1-\sqrt{2}\cos(0)+\ldots$ reaches a *positive* maximum ($+0.293$ at $\theta = 0$). For $n \geq 5$, the analogous maximum is *negative* (e.g. -0.144 for $n = 5$), so no valid domain exists. This is the root of $\mathbb{Z}_3$ uniqueness (Paper VI, Theorem 2).

The universal mass sum rule bridges the gap: it provides a firm prediction for all n without requiring the valid-domain condition.

VII. OPEN PROBLEMS

1. **Physical principle for θ_5.** The transcendental equation (7) gives a unique numerical $\theta_5 \approx 3.70^\circ$. A first-principles derivation from the CDFT vacuum equation of state — analogous to how Paper IV identified the functional form $\Phi_c(\rho_0)$ as the remaining input — is the priority open problem.

2. **Identify the cinquefoil mass spectrum.** Once θ_5 is determined (or bounded), the five mode masses $m_k = M_5 A_k^2$ can be compared to known hadronic states at the ~ 460 MeV energy scale. The total mass $\Sigma_5 = 4603.9$ MeV is a testable sum rule.

3. **Physical meaning of the $3+2$ mode split.** The positive-amplitude modes ($A_k > 0$) and negative-amplitude modes ($A_k < 0$) interact differently with the vacuum background. Whether this split corresponds to a known quantum number (charge, colour, isospin) is an open question.

4. **Higher families $n \geq 7$.** The mass sum rule $\Sigma_n = 2nM_n$ applies to all n. The individual mode spectra and their identification with known or predicted particles require repeating the present analysis for each n.

VIII. CONCLUSION

The universal mass sum rule $\sum_k m_k = 2nM_n$ is derived from two algebraic facts: the $\mathbb{Z}_n$ Fourier identity (which forces $\sum A_k^2 = 2n$ regardless of θ_n) and the Faddeev–Niemi energy scaling $M_n = M_3(n/3)^{3/4}$. Together they give the $n^{7/4}$ mass hierarchy (Eq. (4)), verified for $n = 3$ and predicted for all higher families.

For $\mathbb{Z}_5$ specifically, we prove the $3+2$ mode split (Theorem 2) and derive the transcendental self-consistency equation (7) for θ_5, reporting its solution $\theta_5 \approx 3.70^\circ$ as a hypothesis.

The mass sum $\Sigma_5 = 4603.9$ MeV is a model-level prediction, conditional on the torus-knot hierarchy and the m_e mass-unit calibration.

SUPPLEMENTARY MATERIAL

This manuscript is accompanied by release-archive supplementary material in the canonical Part I tree. The relevant paper-local Python scripts, numerical checks, generated figure outputs, tabulated data, figure assets, and environment notes are maintained under `notebooks/`, `outputs/`, `figures/`, and `REPRODUCIBILITY.md` in `Part_I_Fundamental_Physics/`.

DATA AND CODE AVAILABILITY

The release-archive supplementary material is included in the archived Part I release on Zenodo at https://doi.org/10.5281/zenodo.20250820. The current version DOI is assigned by Zenodo after each GitHub release. Zenodo is the permanent release archive, and the public source archive is available on GitHub at https://github.com/bampita-bico/CDFD-Part-I-Release. The broader public CDFD Runtime is separate reusable software infrastructure; the numerical values reported here are reproduced from this paper's Supplementary Material.

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The author used AI-assisted tools for formatting checks and code-verification support. The scientific claims, interpretations, and final manuscript decisions are the author's responsibility.

The author is indebted to the foundational work of Y. Koide [9] (lepton mass formula), C.A. Brannen [10] (Koide parametrisation), L. Faddeev and A.J. Niemi [7] (knots and particles), and D. Rolfsen [11] (torus knot classification) — whose work underpins the mass sum rule derived here.

DECLARATIONS

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Author Contributions

Steve Bico Mujjabi: conceptualization, methodology, formal analysis, software, validation, visualization, writing–original draft, and writing–review and editing.

Conflicts of Interest

The author declares no conflicts of interest.

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